

Kaili Hampton

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INTRODUCTION

As a Computer Science graduate from Colorado State University, I specialize in full-stack engineering, database architecture, and building scalable software. I am deeply passionate about exploring how complex software systems operate and love writing maintainable code that users love. I am eager to apply my technical foundation and collaborative mindset to build clean, reliable, and high-impact software solutions.

EDUCATION

Colorado State University

Bachelor of Computer Science

Fort Collins, CO

Spring 2024 – Summer 2026

Front Range Community College

Associate of Computer Science

Fort Collins, CO

Fall 2022 – Fall 2023

EXPERIENCE

Colorado State University

Fort Collins, CO

- *Undergraduate Teaching Assistant & Lead TA* *CS150B (FA '24), CS164 (FA '25), CS312 (SP '25, SP '26)*
 - **Course Content Creation & Presentation:** Collaborated in teams of 3–26 to develop and present instructional materials on core programming concepts in Python and Java in CS150B, CS164, and CS312, supporting lecture delivery across multiple courses ranging from 50–250 students.
 - **Supplemental Hours:** Held 3–6 office hours weekly over three semesters to provide 50+ students with one-on-one academic support. Attended course lectures twice weekly to support students and the instructional team.

STEMblazers (formerly She's in Power)

Fort Collins, CO

- *QA/QC Intern* *June 2023 – November 2023*
 - **Database Querying & Cleaning:** Built SQL queries to inspect incoming data for anomalies such as hardware failure, server downtime, and external weather factors that may affect sensor devices.
 - **Data Visualization:** Created visualizations of over 8000 data entries each month using Excel and Python libraries (numpy, SciPy, pandas, etc.). Visualizations were then reviewed and presented to stakeholders during weekly meetings to assess and report estimated total solar generation within the region each month.

Front Range Community College

Fort Collins, CO

- *IT Help Desk* *August 2022 – June 2023*
 - **Ticketing System:** Reviewed, evaluated and assigned 50+ tickets weekly by priority, resolving urgent issues within an hour and lower-priority requests within 12 hours.
 - **Customer Support & Help Desk:** Greeted and assisted in-person students, staff and faculty with IT-related issues. Fielded phone calls for incoming ticket requests and assistance from three locations and offered support for software, hardware, and related configuration in-person, remotely, or via email.

PROJECTS

- **MiniJava to MIPS Compiler:** Built a 4-stage Java compiler (lexing, parsing, type checking, code generation) that translates and converts MiniJava source code into syntactically valid Vapor, Vapor-M, and MIPS.
- **Employee Management System:** Developed in an Agile team of six to build and test an employee management system using Java. Focused heavily on software testing using various methods and tools (JaCoCo, Randoop, Evosuite, PIT) to achieve a minimum of 80% test coverage via ISP/BCC tables and increasing test coverage to 100% using tools such as PMD/CPD, Randoop, and SpotBugs.

TECHNICAL SKILLS

- **Languages & APIs:** Angular, C, C++, C#, Flask, HTML, Java, JavaScript, PHP, Python, SQL, TypeScript
- **Tools:** Bash, Git/GitHub, kanban, Linux, Maven, MS365, Postman, SSH, vim, VSCode, Windows, WinSCP
- **Testing:** Randoop, Spotbugs, PMD/CPD, PIT, JUnit, ISP/BCC, Mockito
- **Databases:** MariaDB, Microsoft Access, MongoDB, MySQL, Postgres